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Representation Stability Under Drifting Stimuli

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Abstract

Stable perception requires maintaining learned representations despite continuous changes in sensory input. We investigate whether self-training can provide a computational mechanism for preserving learned categorical distinctions under gradual stimulus drift.

We first examine this question using a simple theoretical model in which a classifier is required to track a continuously rotating stimulus using its own predictions as learning signals. The model shows that the classifier can follow the changing stimulus while maintaining an approximately constant angular lag.

We then test whether the same principle extends to natural stimuli using face images represented in high-dimensional embedding spaces. A gradual stimulus trajectory was constructed in a two-dimensional PCA space derived from InceptionResNetV1 face embeddings, while classification was performed by a linear perceptron operating on 2,048-dimensional representations extracted from a separate ResNet50 model. After an initial supervised classification was learned, external labels were removed and the classifier was continuously updated using its own predictions as pseudo-labels while the stimuli gradually moved along the trajectory. The classifier successfully tracked the changing stimuli while maintaining the initially learned categorical distinction, even though stimulus drift and classification were defined using representations from different neural network architectures. These results suggest that self-training can support repre-

sentational stability under gradual changes in sensory input by allowing previously learned categorical structure to propagate through a changing stimulus space without continuous external supervision.

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1 Introduction

From birth, humans form internal representations of the external world (Heinen et al. 2023), and in particular, of familiar individuals. For instance, an infant quickly learns to recognize their mother (Bushneil, Sai, and Mullin 1989). Despite continuous changes in the mother’s appearance due to aging and other factors, the child continues to recognize her as the same person. This raises a fundamental question: how can a system maintain a stable representation when its sensory input continuously changes?

In natural environments, sensory input is rarely static. The appearance of objects and individuals changes over time, while their identity and categorical meaning often remain stable. The challenge, therefore, is to explain how a learned categorical distinction can be maintained as the sensory input itself gradually changes.

We hypothesize that self-training may provide a computational mechanism for maintaining such stability. Because natural changes are typically gradual, consecutive observations are likely to correspond to the same object or identity. A system could therefore use its current prediction as a learning signal for subsequent inputs, allowing a previously learned categorical distinction to adapt to gradual changes without requiring continuous external supervision.

In this work, we examine whether self-training can maintain a learned categorical distinction under continuous stimulus drift, first in a simple theoreti-

cal model and then using natural face stimuli represented in high-dimensional neural embedding spaces.

2 Results

2.1 Tracking in the Theoretical Model

We first examined the proposed mechanism using a simple two-dimensional theoretical model. The model consists of two stimuli located at opposite points in a two-dimensional space and a classifier that learns to distinguish between them. Once the classifier reaches a stable state, the stimuli begin to rotate through the space. The classifier is then continuously updated using its own predictions as learning signals.

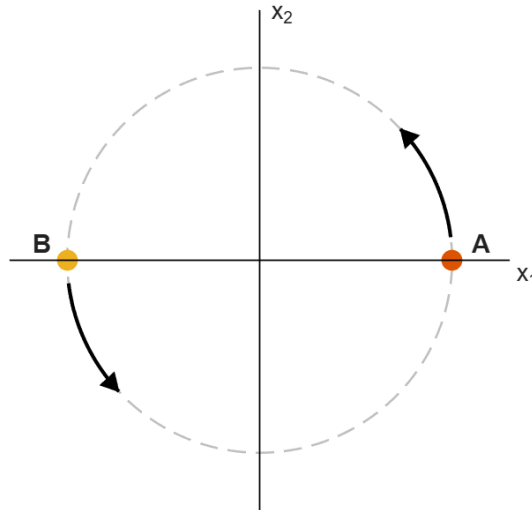


Figure 1: Schematic illustration of the two-dimensional theoretical model. Two opposing stimuli rotate together along the unit circle.

The question is whether the classifier can adapt as the stimuli rotate and continue to preserve the learned distinction between them. The theoretical

analysis shows that the classifier can adapt to the rotating stimulus while maintaining a constant angular lag.

We then simulated the learning process to examine whether this behavior emerges over time. During the simulation, the stimulus was rotated on each trial and the classifier was updated accordingly. We compared the angle of the stimulus with the angle of the classifier’s weight vector throughout the simulation.

The numerical simulation showed that the classifier followed the continuously rotating stimulus while maintaining an approximately constant angular lag, consistent with the theoretical prediction (Loewenstein 2024).

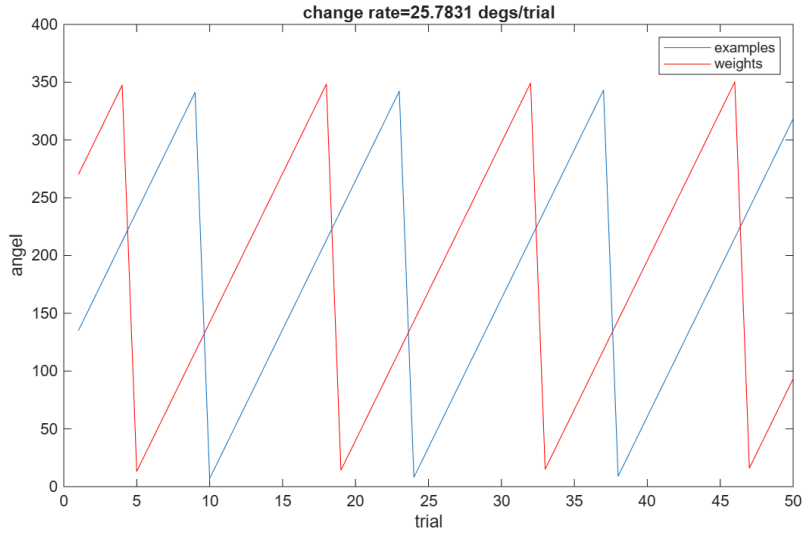


Figure 2: Tracking in the theoretical model. The classifier weight vector follows the continuously rotating stimulus while maintaining an approximately constant angular lag.

2.2 Tracking with Natural Face Stimuli

We tested whether the same tracking mechanism extends from the simple two-dimensional model to natural face stimuli represented in high-dimensional neural embedding spaces. A gradual circular stimulus trajectory was constructed in a two-dimensional PCA space derived from InceptionResNetV1

face embeddings.

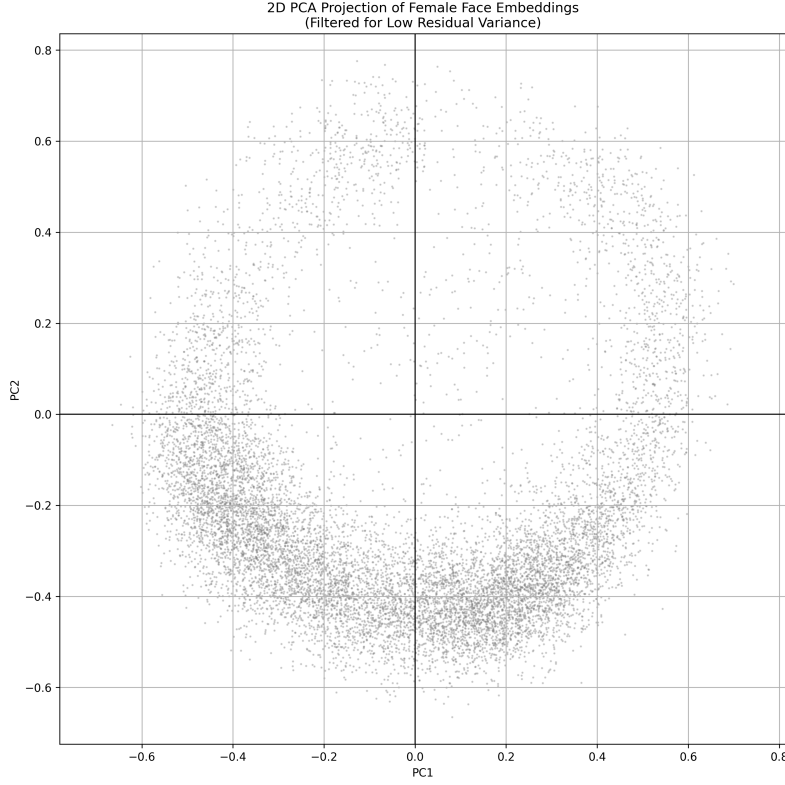


Figure 3: Two-dimensional PCA representation of the face embeddings.
Each point represents a single face image.

Two groups of face images, A and B, were initially selected from opposite positions along this trajectory, and a perceptron classifier was trained to distinguish between them using 2,048-dimensional representations extracted from a separate ResNet50 model.

After this initial supervised classification was learned, external supervision was removed. The centers of the two groups were gradually rotated along the circular trajectory while remaining opposite to one another. At each new position, the face images closest to each center were selected and presented to the classifier. The classifier predicted their categories, and these predictions were then used as pseudo-labels to update the classifier. We then compared the position of the drifting stimuli with the estimated position of the classifier’s decision boundary over time.

The classifier successfully followed the gradually rotating stimuli. The decision boundary rotated together with the stimuli while maintaining a relatively stable angular lag, demonstrating successful tracking even when the stimulus trajectory and classification were based on representations derived from different neural network architectures.

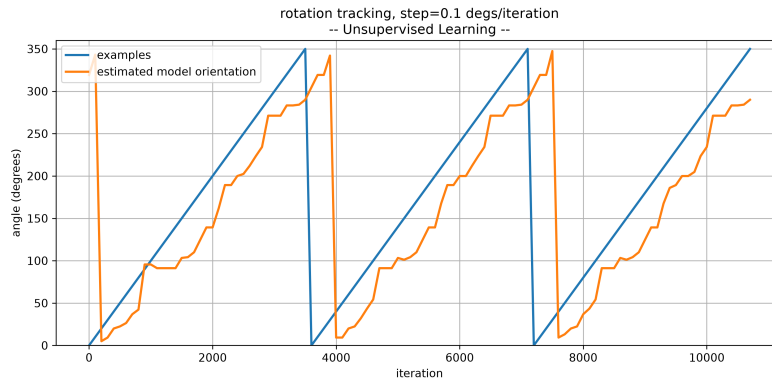


Figure 4: Tracking under gradual stimulus drift using ResNet50 representations. The estimated classifier decision boundary follows the rotating stimulus while maintaining a relatively stable angular lag.

3 Methods

All code used to perform the simulations and experiments is publicly available at [GitHub](#).

3.1 Theoretical Model

We first considered a simplified theoretical model in which a classifier must continuously adapt to a gradually changing input. The model allows us to examine whether a classifier can adapt to continuously changing stimuli while maintaining its learned classification.

3.1.1 Initial Classification Setup

We begin with a stationary two-dimensional classification problem consisting of two examples located at opposite positions,

$$x^1 = (1, 0), \quad x^2 = (-1, 0)$$

with corresponding labels

$$y^1 = 1, \quad y^2 = 0$$

Classification is performed using a logistic function,

$$p = \frac{1}{1 + e^{-z}}$$

where

$$z = wx$$

and

$$w = (w_1, w_2)$$

is the classifier's weight vector. Thus, the orientation of the weight vector determines the orientation of the classification boundary, which is perpendicular to the weight vector.

3.1.2 Learning Dynamics

Learning is mediated by gradient of a cross-entropy loss function with a regularization term:

$$L = - \sum_{j=1}^2 y^j \log(p^j) + (1 - y^j) \log(1 - p^j) + \frac{1}{2} ||w||^2$$

The regularization term prevents the weights from growing without bound and ensures that the learning dynamics converge to a finite, stable solution.

$$\frac{\partial L}{\partial w_i} = (p - y)x_i + w_i$$

Thus, learning updates the weights based on the prediction error, while the regularization term prevents the weights from becoming too large.

3.1.3 Classifier Equilibrium

For stationary stimuli, the learning dynamics are characterized by two fixed points. The first is an unstable fixed point at

$$w_1 = w_2 = 0$$

in which

$$\left\langle \frac{\partial L}{\partial w_i} \right\rangle = 0$$

A second, stable fixed point exists for

$$w_2 = 0$$

and w_1 is the solution to the equation

$$w_1 = \frac{1}{1 + e^{w_1}}$$

for which

$$\forall i \quad \frac{\partial L}{\partial w_i} = 0$$

This stable solution represents the state of the classifier before the stimulus begins to rotate. We next examine how the classifier evolves when the stimulus starts rotating.

3.1.4 Rotating Stimulus

We next allow the input to rotate continuously through the two-dimensional feature space. By symmetry, it is sufficient to consider a single example rotating at a constant angular speed. Its coordinates at time (t) are

$$x_1 = \cos(kt), \quad x_2 = \sin(kt)$$

where k determines the rate at which the stimulus rotates. If the classifier successfully tracks the rotating stimulus, the weights rotate at the same rate as the stimulus while maintaining a constant angular lag. We therefore seek a solution of the form

$$w_1 = r \cos(kt - \psi), \quad w_2 = r \sin(kt - \psi)$$

$$wx = r \cos(\psi)$$

where r is the magnitude of the weight vector and ψ is the angular lag between the rotating stimulus and the classifier. To determine how the weights must change in order to track the rotating stimulus, we consider one learning step, starting at $t = 0$. At this point, the weight vector is

$$w(0) = \begin{pmatrix} r \cos(-\psi) \\ r \sin(-\psi) \end{pmatrix}$$

After one step, the stimulus has rotated by an angle (k). If the weights successfully track the stimulus while maintaining the same angular lag (ψ), the weight vector becomes

$$w(1) = \begin{pmatrix} r \cos(k - \psi) \\ r \sin(k - \psi) \end{pmatrix}$$

Therefore, the change in the weights required to maintain tracking is

$$\Delta w = w(1) - w(0) = \begin{pmatrix} r \cos(k - \psi) - r \cos(-\psi) \\ r \sin(k - \psi) - r \sin(-\psi) \end{pmatrix}$$

We now consider the change in the weights produced by the learning rule. According to gradient descent, the weights are updated by

$$\Delta w = -\eta \nabla_w L$$

where η is the learning rate. As defined above, the gradient of the loss is

$$\nabla L = (p - y)x + w$$

we look at

$$x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y = 1$$

and

$$wx = r \cos(\psi)$$

therefore

$$p = \frac{1}{1 + e^{-r \cos(\psi)}}$$

so the change in the weights produced by learning during one step is

$$\Delta w = -\eta \left(\left(\frac{1}{1 + e^{-r \cos(\psi)}} - 1 \right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} r \cos(-\psi) \\ r \sin(-\psi) \end{pmatrix} \right)$$

For the classifier to successfully track the rotating stimulus, this change must equal the weight change required to maintain the rotation derived above.

$$\Delta w = \begin{pmatrix} r \cos(k - \psi) - r \cos(-\psi) \\ r \sin(k - \psi) - r \sin(-\psi) \end{pmatrix} = -\eta \left(\left(\frac{1}{1 + e^{-r \cos(\psi)}} - 1 \right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} r \cos(-\psi) \\ r \sin(-\psi) \end{pmatrix} \right)$$

From the second component of the equation, we obtain the relationship between the rotation rate k and the angular lag ψ :

$$\sin(k - \psi) + \sin(\psi) = \eta \sin(\psi)$$

which gives

$$\psi = \arctan\left(\frac{\sin(k)}{\eta - 1 + \cos(k)}\right)$$

So when $k = 0$, it gives $\psi = 0$. Thus, in the stationary case, there is no angular lag between the stimulus and the classifier. To compute r we use the first equation:

$$r \cos(k - \psi) - r \cos(-\psi) = -\eta \left(\left(\frac{1}{1 + e^{-r \cos(\psi)}} - 1 \right) + r \cos(-\psi) \right)$$

which should be solved numerically.

The theoretical model therefore describes a possible state in which the classifier follows the rotating stimulus with a constant angular lag. We next examined whether this behavior emerges in a numerical simulation of the learning dynamics.

3.1.5 Numerical Simulation

The model was simulated for 50 trials. On each trial, the stimulus rotated by 0.45 radians ($\approx 25.8^\circ$), and the classifier was updated once using gradient descent with a learning rate of 0.1.

The label used for learning was determined by the classifier's current prediction. Thus, the classifier continuously learned from its own classifications as the stimulus gradually rotated.

During the simulation, we measured the angles of the stimulus and the classifier weight vector to examine whether the classifier could follow the rotat-

ing stimulus while maintaining its classification.

3.2 From Theory to Natural Stimuli

To examine whether the same principle observed in the theoretical model extends to natural stimuli, we tested it using natural face images represented in high-dimensional embedding spaces and a simple perceptron classifier.

Face images were obtained from the CelebA dataset (Liu et al. 2015), and only images labeled as female were retained for the analysis. Face representations were initially extracted using an InceptionResNetV1 model from the facenet-pytorch library (Esler 2019), pretrained on the VGGFace2 dataset (Cao et al. 2017), yielding a 512-dimensional embedding for each image. This representation follows the general FaceNet paradigm of representing faces in a continuous embedding space where images of the same identity are mapped closer together and images of different identities are mapped farther apart (Schroff, Kalenichenko, and Philbin 2015).

Principal component analysis (PCA) was applied to the 512-dimensional embeddings. We first determined the number of principal components required to explain at least 95% of the total variance. The embeddings were then projected onto these components. To obtain a subset of images that lay close to the two-dimensional subspace defined by PC1 and PC2, we computed, for each image, the squared Euclidean distance from the origin in the residual PCA space spanning PC3 through the final component required to reach 95% explained variance. We retained the 10% of images with the smallest residual distance. These images were then represented by their coordinates on PC1 and PC2. Within this filtered two-dimensional space, a circular trajectory was defined to guide the selection of face images across successive iterations, thereby producing a gradual shift in the stimuli pre-

sented to the classifier.

Classification was performed using representations extracted from a separate ResNet50 model trained on VGGFace2, allowing the stimulus trajectory and the classifier to operate on representations derived from two distinct neural network architectures.

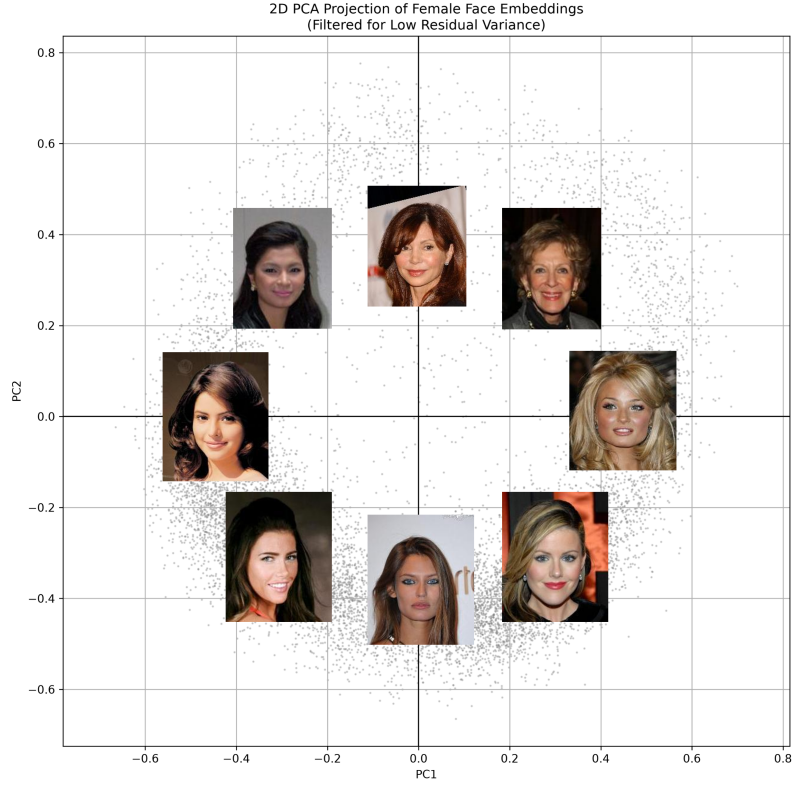


Figure 5: Gradual change of face stimuli along the circular trajectory in PCA space

3.3 Initial Classification

Before stimulus drift began, two groups of face images, A and B, were selected from opposite regions of the trajectory, centered at 0° and 180° , with 1,000 images in each group. A simple perceptron classifier was trained under external supervision to distinguish between the two groups. Training was performed for 100 epochs using a cross-entropy loss, with a learning rate of 0.001 and a weight decay of 1. The classifier operated on 2,048-

dimensional face representations extracted using a ResNet50 model pre-trained on VGGFace2 and consisted of a single linear layer mapping these representations to two output units, corresponding to categories A and B. This initial supervised training established the categorical distinction that the model was subsequently required to maintain as the stimuli gradually changed.

```
self.classifier = nn.Linear(
    EMBEDDING_DIM,
    2
)
```

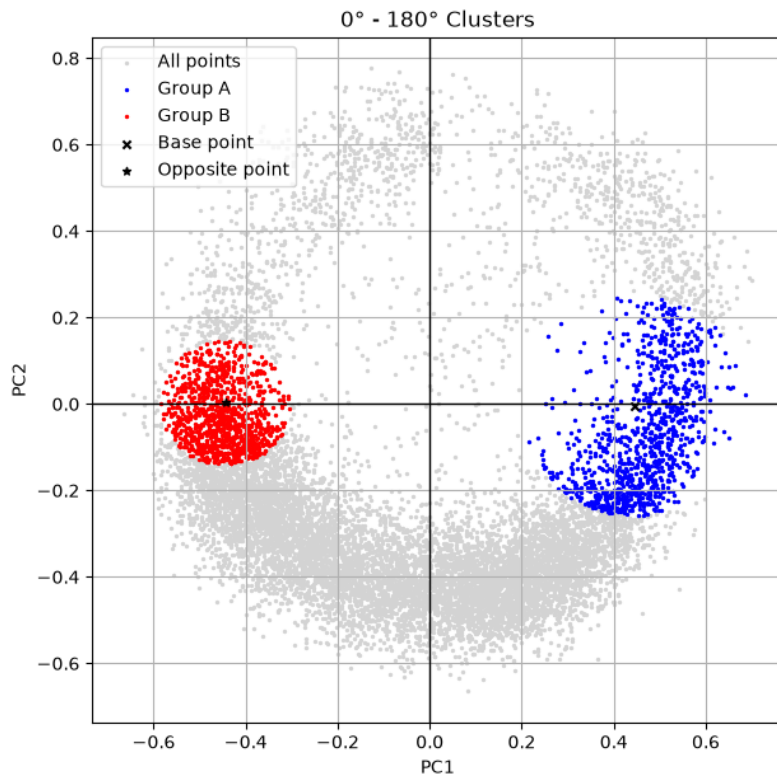


Figure 6: Initial A and B clusters in the InceptionResNetV1 PCA space

3.4 Self-Training Under Stimulus Drift

After the initial classification was learned, external supervision was removed and training continued using self-generated pseudo-labels. Two opposite

points in the InceptionResNetV1 PCA space, initially positioned at 0° and 180° , defined the centers of clusters A and B. At each iteration, the 100 images closest to each cluster center in the PCA space were selected. The corresponding 2,048-dimensional ResNet50 representations of these images were then provided as input to the current perceptron classifier. The classifier predicted category A or B for each image, and these predictions were subsequently used as pseudo-labels for training. The perceptron was then trained for one epoch on the selected images using these pseudo-labels, with a cross-entropy loss and the AdamW optimizer, using a learning rate of 0.001 and a weight decay of 1.

Following each training iteration, both cluster centers were rotated counterclockwise by 0.1° in the InceptionResNetV1 PCA space, while remaining 180° apart. The selection, prediction, and self-training procedure was then repeated at the new positions. This process was performed for 10,800 iterations, corresponding to three complete 360° rotations through the PCA space.

3.5 Measuring Tracking

The position of the drifting stimuli was described by their angle along the circular trajectory. To estimate the position of the classifier, we identified the point along the trajectory at which its predictions switched between categories A and B, and used the corresponding angle as an estimate of the classifier’s decision boundary angle. Tracking was then evaluated by comparing this angle with the stimulus angle over successive iterations. Successful tracking was defined as the classifier following the rotating stimuli while maintaining a relatively stable angular lag between the stimulus and the estimated decision boundary.

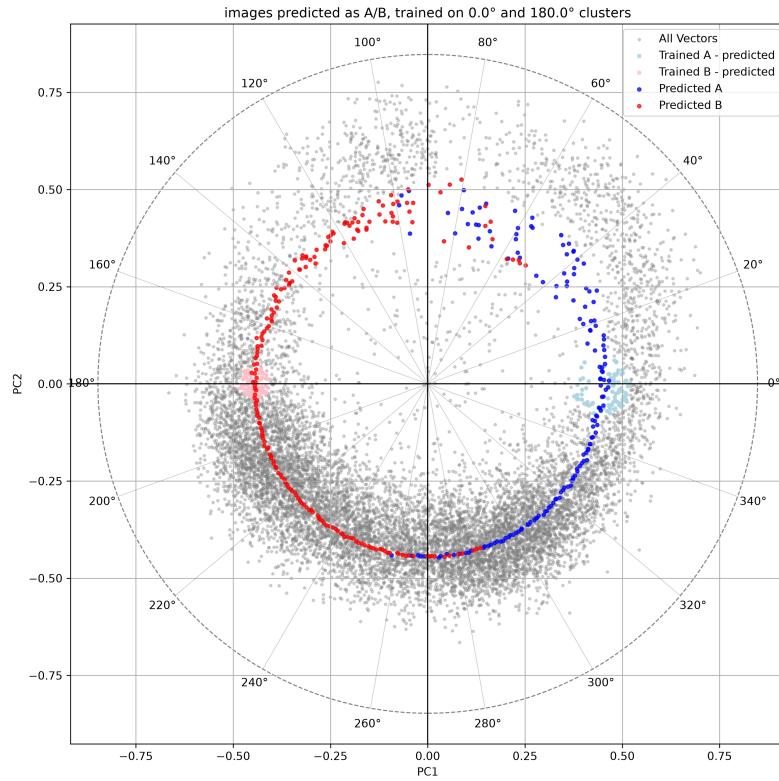


Figure 7: Estimation of the classifier decision boundary along the stimulus trajectory. Colors indicate the classifier's predicted category for face images sampled along the circular trajectory. The angle at which predictions switch between categories A and B was used to estimate the classifier's decision boundary angle.

4 Discussion

4.1 Main Findings

The theoretical model demonstrates that a classifier can follow continuously changing stimuli through self-training, without requiring continuous external supervision. During successful tracking, the classifier follows the changing stimuli while maintaining a constant angular lag. We found that the same principle extends to natural face images represented in high-dimensional embedding spaces. Importantly, successful tracking was observed when the stimulus trajectory was defined using InceptionResNetV1 representations while classification was performed using representations extracted from ResNet50.

4.2 Stability Requires Adaptation

When stimulus representations gradually change, a fixed decision boundary can no longer preserve the initially learned categorical distinction. Maintaining this distinction therefore requires the decision boundary to adapt together with the changing stimuli. Our results show that self-training can provide such a mechanism: by using its own predictions as pseudo-labels, the classifier can continuously update its decision boundary without receiving new external labels.

4.3 The Role of Temporal Continuity

The proposed mechanism relies on the gradual nature of stimulus change. When changes between consecutive stimuli are sufficiently small, the classifier can use its predictions as pseudo-labels to adapt to the gradually changing stimuli. In this way, a previously learned categorical distinction can be propagated through a continuously changing stimulus space without continuous external supervision. During successful tracking, the classifier adapts continuously to the changing stimuli while maintaining an approximately constant angular lag behind them.

4.4 Generalization beyond the theoretical model

In the theoretical model, the rotating stimuli and the classifier are defined within the same two-dimensional space: the classifier directly receives the coordinates of the rotating stimulus as input. In the face experiment, however, the stimulus drift is defined in the PCA space derived from Inception-ResNetV1 embeddings, whereas the classifier receives 2,048-dimensional representations extracted from ResNet50. Despite this separation, the classifier successfully tracks the gradually changing stimuli through self-training. This indicates that successful tracking does not require the classifier to operate directly in the space in which the stimulus trajectory is defined.

4.5 Limitations and future directions

In the current experiment, stimulus drift is constructed using face images of different individuals rather than gradual changes in the same individual. Therefore, the trajectory represents gradual change in the representational space, but not the temporal evolution of a single identity. A future experi-

ment could follow different images of the same individual as their appearance changes over time, for example due to aging. This would provide a more direct test of whether self-training can maintain a stable identity representation as the sensory input associated with the same person gradually changes.

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